SATVIK SRIRAM

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Education

University of California, San Diego – B.S. Computer Science **Graduated: June 2025**
Overall GPA: 3.94/4.00

Technical Skills

Coding Languages: Java, Python, C, C++, HTML, CSS, JavaScript, TypeScript, SQL, Unix, Bash and Shell Scripting

Libraries: React, Node.js, Vite, Scikit-learn, TensorFlow, OpenCV, PyTorch, keras, Matplotlib, NLTK, Selenium, Puppeteer

Tools: Git/Github, Retrium, Miro, Figma, CI/CD, Postman

Certifications

AWS Certified Solutions Architect - Associate | Amazon Web Services (AWS) **Date: Oct 2025**
AWS Certified Cloud Practitioner | Amazon Web Services (AWS) **Date: Sept 2024**

Professional Experience

Cohort Lead/Teaching Assistant ***Berkeley Coding Academy*** **Date: 2023 - Present**

- Designed and delivered structured machine learning course materials while collaborating with a team of 5+ tutors to administer course materials to students of various skill levels.
- Simplified complex concepts in machine learning, data processing, and neural networks enabling students aged 10+ to grasp foundational principles and apply them to practical projects.
- Provided personalized 1-1 sessions, developing custom lesson plans to 6+ students based on individual needs.

Software Engineer Intern ***BlueCloak*** **Date: 2021**

- Worked with a team of 7 interns and 20+ current employees to construct a company-assigned project.
- Engineered a real-time internet traffic generator to simulate realistic data transfers between users and data servers to prepare for possible cyber-attacks.
- Utilized Python with NLTK and SpaCy integration to simulate realistic internet sessions with artificial users.

Projects

Pokemon TCG Price Scanner **Date: 2025 - Present**

- Engineering a full-stack real-time Pokémon card price retrieval app using React, FastAPI, WebSockets, and TensorFlow for live detection, matching, and pricing from webcam input.
- Deployed a custom-trained YOLO-based segmentation model that achieved 99.9% precision with OpenCV and NumPy for efficient in-frame card detection, perspective correction, and visual annotation in live video streams.
- Implemented a server-side price caching system (TTLCache) and RESTful API powered by FastAPI to enable low-latency batch price lookups from the Pokémon TCG API; reducing API load by 70% and response latency by 60%.
- Built a scalable WebSocket backend in Python for real-time frame ingestion and model inference, with frontend integration in React/Vite for smooth user experience across devices with a detection-to-price delay of 100ms

HobbyList **Date: 2025 - Present**

- Developing a full-stack social web hosted using React and TypeScript and backed by Java, Spring Boot, Spring Security, JPA/Hibernate, and a PostgreSQL database, enabling verified user accounts, hobby-tagged photo uploads via AWS S3.
- Engineered RESTful API and optimized data access through JPA and JDBC, improving API response time by 45% via query batching, connection pooling, and presigned S3 URL handling for seamless media uploads.
- Implemented a secure email verification system using Spring Boot and JavaMail, ensuring account authenticity with tokenized verification links and expiration-based validation.

Root Journal **Date: 2024**

- Led a team of 10 developers in creating a journaling web application using Agile methodologies and 10-day sprints.
- Detailed timelines, design sketches, and wireframes through Miro and Figma while planning meetings and goals.
- Utilized Quilljs in order to have built-in document editing features such as text size, font, color, and bullet lists.
- Leveraged CSS and JavaScript to visually animate the growth of a root over a year with low performance impact while storing 1,000s of journal entries per user.

Stock Market Predictions Based on News Headlines **Date: 2024**

- Worked with a team of 6 AI engineers and data scientists to conduct a study on AI's ability to predict stock market fluctuations based on news headlines.
- Capitalized on the NLTK library to preprocess thousands of news headlines and transform them into word embeddings.
- Modified a transformer encoder to achieve 74.3% training accuracy and a 61.5% testing accuracy on stock predictions.